Abhishek Kumar Singh

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EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur **2025-2029 (Expected)**
B.Tech, Electronics & Communication Engineering (IoT) | First-year CGPA: 8.4/10

Kendriya Vidyalaya, IIT Kanpur Campus
Senior School Certificate Examination (XII): 89% **2024**
Secondary School Examination (X): 89.4% **2022**

PROJECTS

AeroNode | Adaptive Edge-AI Inference Runtime **Oct 2026**
github.com/abhishekkr10/AeroNode | Tech Stack: C++17, ONNX Runtime

- Engineered staged C++17 inference with persistent ONNX Runtime sessions and confidence-based early exits, skipping unnecessary stages while checking predictions against reference execution.
- Measured 25.2% lower mean latency (2.1052 ms vs 2.8139 ms) at 72.6857% vs 72.9400% accuracy on 1,966 images held out from threshold fitting (Windows CPU, batch 1, one thread, three repetitions).

CausalBench | Reproducible Controller-Failure Lab **Nov 2026 - Present**
github.com/abhishekkr10/CausalBench | Tech Stack: Python

- Replayed a delayed sensor reading that triggered a pulse in the baseline controller. The guarded controller rejected the stale reading and held its output in simulation.
- Automated failure minimization, regression checks and source/evidence exports; documented validation with 93 Python tests and 13 browser workflows.

SafeTrack | Signed IoT Shipment Monitoring | Team project **Jul 2026**
github.com/inamansingh2401/tamperproject (team repo) | Tech Stack: ESP32-S3, DHT11, MPU6050, ATECC508A, Python, SQLite, FastAPI, PostgreSQL

- Designed the end-to-end architecture and wrote ESP32-S3 firmware for DHT11, MPU6050 and light sensing, with ATECC508A-signed, hash-linked events for tamper-evident shipment records.
- Built the Python gateway with durable SQLite outbox, FastAPI backend and PostgreSQL storage (migrated from Render to Supabase); contributed to Admin/User dashboard design.

JOCKY | Forensics Analysis Platform **Sep 2026**
github.com/abhishekkr10/jocky-forensics | Tech Stack: Rust

- Compiled investigation scripts into deterministic analysis plans and verified signed evidence checkpoints in Rust; provided a synthetic public demo.

TECHNICAL SKILLS

Embedded/Hardware: C, C++17, ESP32-S3, I2C, UART, MPU6050, DHT11, ATECC508A
ML/Inference: Python, PyTorch, ONNX, ONNX Runtime
Backend/Tools: FastAPI, PostgreSQL, SQLite, ZeroMQ, React, TS/JS, Rust, CMake, Git/GitHub

ACHIEVEMENTS

- Finalist, Techfest IIT Bombay - The Wrecking Ball Challenge (38 teams qualified)
- Ranked 9th of 180 teams in the Smart India Hackathon 2026 internal round (MMMUT) with SafeTrack